

Case 05: Design and Development of a Shoulder Replacement Prosthesis Using SolidWorks

1. Introduction

Shoulder replacement arthroplasty is a surgical procedure used to replace damaged components of the shoulder joint caused by osteoarthritis, rheumatoid arthritis, trauma, avascular necrosis, or severe fractures. The procedure aims to restore joint function, reduce pain, and improve range of motion.

The shoulder joint consists of the humeral head and glenoid cavity. In shoulder replacement surgery, the damaged humeral head is replaced with a prosthetic component designed to mimic the natural anatomy of the shoulder joint.

2. Need for Shoulder Replacement Implant

- Severe shoulder arthritis
- Complex proximal humerus fractures
- Avascular necrosis
- Joint degeneration
- Chronic shoulder pain
- Reduced range of motion

3. Objective

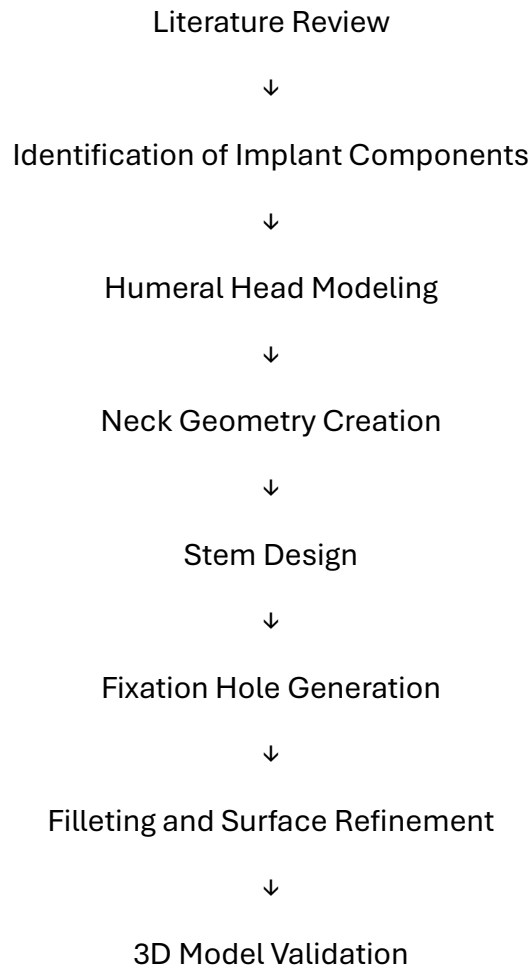
- To design a shoulder replacement prosthesis using SolidWorks.
- To understand the geometry and design principles of shoulder implants.
- To develop a CAD model suitable for orthopedic applications.
- To study implant design and fixation features.

4. Software Used

Component	Software
CAD Modeling	SolidWorks
Design Validation	SolidWorks
Assembly Visualization	SolidWorks

STL Generation	SolidWorks
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5. Methodology



6. Design Features

Feature	Function
Humeral Head	Replaces damaged articular surface
Anatomical Neck	Maintains joint biomechanics
Intramedullary Stem	Provides fixation within humerus
Fixation Holes	Additional implant stability
Tapered Stem	Facilitates insertion into medullary canal
Smooth Surface Finish	Reduces tissue irritation

7. Materials Used

Component			Material	Reason for Selection		
Humeral Stem			Ti-6Al-4V ELI	High strength and biocompatibility		
Humeral Head			CoCrMo Alloy	Excellent wear resistance		
Fixation	Screws	(if required)	Titanium Alloy	Corrosion resistance	and compatibility	

8. Design Specifications of Shoulder Prosthesis

Parameter	Specification
Implant Type	Shoulder Replacement Prosthesis
Design Method	CAD-Based Anatomical Modeling
Stem Material	Ti-6Al-4V ELI
Head Material	CoCrMo Alloy
Stem Geometry	Tapered Intramedullary Stem
Fixation Method	Press Fit / Screw Fixation
Manufacturing Method	CNC Machining / Additive Manufacturing
File Format	STL

9. Importance of the Implant

Clinical Importance	Benefit
Pain Relief	Reduces chronic shoulder pain
Joint Function Restoration	Improves arm movement
Range of Motion Improvement	Enhances daily activities
Load Distribution	Restores shoulder biomechanics
Long-Term Stability	Improves implant lifespan
Improved Quality of Life	Better functional outcome

10. Additive Manufacturing Compatibility

Parameter	Specification
Material	Ti-6Al-4V ELI
Manufacturing Technology	Selective Laser Melting (SLM)
Layer Thickness	30–60 μm
Post Processing	Machining and Polishing
Sterilization	Medical Grade Sterilization

